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SOFTWARE REQUIREMENTS SPECIFICATION

ClinicQ

Clinic Appointment and Live Queue System

including user stories

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1. Introduction

> **In plain words:** This section says what ClinicQ is for, who the document is for, what is and is not included, and what could go wrong.

1.1 Purpose

This Software Requirements Specification (SRS) describes what the ClinicQ system shall do and how well it shall do it. ClinicQ is a web-based system that lets patients book clinic appointments online and follow their place in a live queue on the day of the visit. It also gives receptionists, doctors and the clinic manager the tools to run the day smoothly.

The SRS is the agreed starting point for design, coding, testing and project tracking. It states requirements for **all stakeholders** (patients, clinic staff and the clinic owner), not only for the development team.

1.2 Intended Audience

Audience	Why they read this SRS
Developers	To know exactly what to build and which rules the code must enforce.
Testers / QA	To write test cases from the confirmation criteria, business rules and measurable targets.
UI/UX designers	To design screens for each user class and check them against Nielsen's heuristics.
Project manager / team lead	To plan sprints, track progress and control scope.
Course instructor and examiners	To assess the completeness and quality of the requirements work.
Clinic stakeholders (doctor, receptionist, manager)	To confirm the system matches how the clinic really works.
Future maintainers	To understand why the system behaves as it does when it is changed later.

1.3 Intended Use

- **Baseline for design:** the Software Design Specification (SDS: use case diagram, data flow diagram and state diagram) is derived from this document.
- **Baseline for testing:** every user story has confirmation criteria that become acceptance tests; every business rule that is built gets at least one unit test.
- **Planning tool:** stories are prioritised (MoSCoW) and estimated (story points) so the team can plan one-week sprints.
- **Change control:** after approval, a requirement changes only through a recorded change request and a new version number.

1.4 Product Scope

Product name: ClinicQ (working name). **Type:** a new, stand-alone web application for a single clinic with several doctors.

What it does. Patients find a doctor, book, change or cancel an appointment, and check in. During the day the system keeps a live queue for each doctor, estimates waiting time, and tells patients when their turn is near. Doctors call patients and record visit notes and prescriptions. Receptionists handle phone bookings, walk-ins and emergencies. The clinic manager manages staff, rules and reports.

Benefits

- **Patients:** less uncertainty and wasted time; booking at any hour; records in one place.
- **Receptionists:** fewer phone calls, no double bookings, a clear queue instead of a paper register.
- **Doctors:** a clear view of the day, fewer interruptions, quick access to history.
- **Clinic owner:** fewer no-shows, better use of doctors' time, simple reports.

Objectives and measurable goals

ID	Objective	How success is measured
OBJ-1	Online booking without double bookings	100 simultaneous booking attempts for one slot produce exactly one booking (REL-02).
OBJ-2	Patients know where they stand in the queue	Position and waiting-time estimate update within 5 seconds of any queue change (PERF-03).
OBJ-3	Trustworthy waiting-time estimates	In simulated queue runs of at least 200 patients, at least 90% of estimates fall within $\pm$10 minutes of the actual wait (PERF-05).
OBJ-4	Fast front-desk work	A walk-in is registered and queued in 30 seconds or less (USAB-03).
OBJ-5	Safe handling of urgent cases	An emergency flag always places the patient first and is logged (SAFE-02).
OBJ-6	Quality that can be proven	Every business rule that is built has at least one automated unit test (MAIN-02).

Scope boundaries

In scope (this release)	Out of scope ("Won't" in this release)
• Single clinic, multiple doctors • Patient, doctor, receptionist and admin roles • Doctor schedules, leave and clinic rules • Online and reception booking, cancel, reschedule • Check-in, walk-ins, live queue, emergency flag • Visit notes and simple prescriptions • SMS, email and in-app notifications • Patient self-service: profile, mobile-number change, data copy and deletion request • Reports and CSV export • Bangla and English interface	• Online payments and billing • Video or phone consultation • Multiple clinic branches • Pharmacy, stock and lab integration • Insurance claims • Integration with hospital or national health records (for example, HL7 FHIR) • A native mobile app (the responsive web app is used on phones) • Diagnosis or medical advice of any kind

The scope is deliberately limited so that a student team can finish, test and demonstrate it well within the semester.

1.5 Risk Definitions

A **risk** here is something that could go wrong when ClinicQ is used and cause harm, loss or unfair treatment. Defining risks early lets us write requirements that prevent or reduce them. (Project risks such as schedule or team problems belong to the project plan, covered in the course unit on risk management.)

Rating: Likelihood (1 = low, 2 = medium, 3 = high) × Impact (1 = minor, 2 = serious, 3 = severe). Score 6–9 = High, 3–5 = Medium, 1–2 = Low. Any risk that could harm a patient's health is treated as High whatever its score.

ID	What could go wrong	Consequence	Rating	Mitigated by
R-01	Two patients are booked into the same slot.	Disputes, wasted trips, lost trust.	$2 \times 3 = 6$ High	BR-01, REL-02, BKG-03
R-02	The queue order is unclear or seems unfair.	Arguments, complaints, staff stress.	$2 \times 2 = 4$ Medium	BR-08, BR-10, QUE-04, QUE-06
R-03	An urgent patient waits behind routine patients.	Possible harm to the patient's health.	$1 \times 3 = 3$ High (patient safety)	BR-09, QUE-09, SAFE-02
R-04	Someone sees patient data they should not.	Privacy breach, legal and reputation damage.	$2 \times 3 = 6$ High	SEC-01, SEC-06, PRIV-05, BR-15, VIS-02, AUTH-07, AUTH-08, AUTH-09
R-05	Appointment or visit data is lost.	Lost history, scheduling chaos.	$1 \times 3 = 3$ Medium	REL-03, SYS-03
R-06	A confirmation or alert is not delivered.	Missed appointments, confusion.	$3 \times 2 = 6$ High	REL-04, NOT-01, QUE-04 (in-app fallback)
R-07	The doctor is on leave but patients still arrive.	Wasted trips, unhappy patients.	$2 \times 2 = 4$ Medium	BR-13, SCH-03, NOT-03
R-08	The waiting-time estimate is badly wrong.	Patients lose trust in the system.	$3 \times 2 = 6$ High	BR-12 (range, real averages), QUE-13, PERF-05
R-09	The system is down during clinic hours.	Clinic cannot manage bookings or queue.	$2 \times 3 = 6$ High	REL-01, SYS-04 (printed schedule, Must)
R-10	Patients treat the system as medical advice.	Misuse; liability.	$1 \times 2 = 2$ Low	SAFE-01
R-11	People hoard slots or repeatedly miss appointments.	Fewer slots for others; idle doctors.	$3 \times 2 = 6$ High	BR-06, BR-07, BR-16, BKG-07, QUE-14, ADM-04

1.6 Definitions and Acronyms

Term	Meaning
Appointment	A reserved slot with a doctor for one patient.
Slot	A fixed block of time in a doctor's schedule (default 15 minutes) that can hold one appointment.
Walk-in	A patient who arrives without an appointment.
Queue	The ordered list of patients waiting for one doctor on one day.
Token	A short number given at check-in (for example D2-014) so patients can be called without using their names.
Check-in	Recording that a patient has arrived; it puts them in the queue.
Grace period	Time after the slot start during which a late patient still keeps appointment priority (default 15 minutes).
Late arrival	A patient with an appointment who checks in after the grace period but before the no-show limit. They keep the appointment and join the queue from their check-in time; they do not count as a no-show.

Term	Meaning
No-show limit	Time after the slot start (default 60 minutes) after which a booked patient who has not checked in is marked as a no-show.
Effective time	The time used to order the queue: slot start for on-time appointment patients; check-in time for walk-ins and late arrivals.
No-show	A booked patient who did not check in by the no-show limit.
Emergency flag	A mark set by a receptionist or doctor that moves a patient to the front of the queue.
EWT	Estimated waiting time: the approximate time until a patient is called.
Dependent	A person (child or parent) whose appointments are managed from another patient's account.
Visit record	The doctor's notes, advice and prescription from one consultation.
Audit log	A history of who did what and when, which cannot be edited in the application.
Epic	A group of related user stories.
User story	A short requirement from the user's point of view: "As a . . . , I want . . . , so that . . . ".
Confirmation	The Given / When / Then checks that prove a story is done (also called acceptance criteria).
MoSCoW	Priority scale: Must, Should, Could, Won't.
Story point	A relative size estimate of the effort a story needs.
Sprint	A fixed one-week work period with a set of stories to finish.
SRS / SDS	Software Requirements Specification (this document) / Software Design Specification.
DFD	Data flow diagram: shows how data moves between people, processes and storage.
RBAC	Role-based access control: what you can do depends on your role.
OTP	One-time password sent by SMS to confirm a mobile number.
MVC / ECB / CQRS	Architecture patterns: Model-View-Controller; Entity-Control-Boundary; Command Query Responsibility Segregation.
WCAG	Web Content Accessibility Guidelines, the international standard for accessible websites.
TLS / HTTPS	Encryption that protects data travelling between the user and the server.
RPO / RTO	Recovery point objective (most data we can lose) / recovery time objective (longest downtime we accept).
p95	The 95th percentile: 95% of measurements are at or below this value.
CI	Continuous integration: automatic building and testing of every code change.

1.7 References

- ISO/IEC/IEEE 29148:2018, *Systems and software engineering — Life cycle processes — Requirements engineering*.
- IEEE Std 830-1998, *Recommended Practice for Software Requirements Specifications* (superseded by 29148, used here for section layout).
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- W3C (2018). *Web Content Accessibility Guidelines (WCAG) 2.1*.
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- CSE 451 course materials: Course Outline, Project Assessment, “How to Write an SRS”, Lecture 1 (Software and Software Engineering) and Lecture 2 (Software Development Process Models), Prof. MM Mahbubul Syeed, IUB.

1.8 Document Conventions

Convention	Meaning
Story ID	EPIC-NN, for example BKG-03 (BKG = booking epic, story number 03).
Other IDs	BR = business rule · PERF, REL, SEC, PRIV, USAB, COMP, MAIN, SAFE, LEGAL = non-functional requirement groups · R = risk · OBJ = objective · A = assumption · C = constraint.
Priority (MoSCoW)	**Must:** needed for the complete core demo (book, check in, queue, consult). **Should:** important, built if time allows. **Could:** nice to have. **Won’t:** deliberately left out of this release (Section 1.4).
Dependencies	The confirmation criteria of a Must story never depend on a Should or Could story (checked in Appendix A.5).
Single source of truth	A number or rule is written in one place only: in a business rule (BR) or a non-functional requirement. Stories point to it and may use its default value as an example.
Story points	Relative effort using 1, 2, 3, 5, 8. A 1-point story is a few hours; an 8-point story is the biggest we allow (split anything larger). Estimates will be re-checked after Sprint 1.
Story format	As a [role], I want [goal], so that [benefit]. Each story has a Confirmation: Given [situation], when [action], then [result].
Story quality (INVEST)	Independent, Negotiable, Valuable, Estimable, Small, Testable.
Wording	In non-functional requirements, “shall” means mandatory and “should” means recommended.
Verification	T = test · I = inspection or review · D = demonstration · A = analysis.
Time and language	All times are Asia/Dhaka (UTC+6). The document is in English; the product supports Bangla and English.

2. Overall Description

> **In plain words:** This section describes the product, who uses it, what they need, where it runs, and what we are assuming.

What we are building and why

ClinicQ is a **new product** (it does not replace or extend an existing software system). It is for a single clinic with several visiting doctors. Many clinics today rely on a paper serial register, phone calls and a crowded waiting room. Patients cannot tell when they will be seen, receptionists struggle with double bookings and walk-ins, and doctors have no clear view of their day. ClinicQ replaces this with online booking, a live queue and simple visit records.

Main functions

- **Accounts and Access** — Who can enter the system, and what each person is allowed to do.
- **Doctors and Schedules** — How doctors and the clinic define the times when appointments can be booked.
- **Appointment Booking** — Finding a doctor, choosing a time, and changing or cancelling it.
- **Live Queue** — What happens on the day: check-in, who is next, and how long people wait.
- **Consultations and Records** — What the doctor records during and after the visit.
- **Notifications** — Messages that keep patients informed without calling the clinic.
- **Reports and Oversight** — Helping the clinic manager see how the clinic is running.
- **Cross-cutting Needs** — Needs that touch the whole system: language, phone use, backups, and privacy.

A patient's visit journey

Stage	What the patient does	What ClinicQ provides	Epic
1. Find	Looks for a suitable doctor	Search by name or specialty, doctor profiles	BKG
2. Book	Chooses a free slot	Live availability, instant confirmation, no double booking	BKG, NOT
3. Prepare	Waits for the day	Reminders, easy cancel or reschedule	NOT, BKG
4. Arrive	Reaches the clinic	Check-in by receptionist or QR slip; token issued	QUE
5. Wait	Sits or steps out	Live position, estimate, alerts, public display	QUE, NOT
6. Consult	Sees the doctor	Doctor calls next patient; notes and prescription recorded	QUE, VIS
7. Follow up	Leaves and returns later	Records, prescription, follow-up booking	VIS, BKG

2.1 User Classes and Characteristics

User class	Type	Characteristics	Main tasks
Patient (and guardian of dependents)	Primary	Wide range of ages and digital skills. Mostly uses a smartphone. Visits are occasional. May book for family members. Needs very simple screens.	Register, search, book, cancel or reschedule, check in, watch the queue, read own records.
Receptionist	Primary	Uses the system all day on a PC. Works under time pressure, with phone calls and walk-ins at the same time. Needs speed and few clicks.	Phone bookings, patient registration, check-in, walk-ins, emergency flag, queue fixes, printing schedules.
Doctor	Primary	Busy and interrupted often. Uses a PC or tablet in the consulting room. Wants a clear queue and fast note-taking.	Set schedule and leave, view queue, call next, record notes and prescriptions, follow-ups.
Clinic admin (manager)	Secondary	Uses the system now and then. Responsible for staff, rules and results. Comfortable with dashboards and reports.	Staff accounts, clinic rules, holidays, reports, audit log, backups.
Clinic owner	Buyer / sponsor (not a regular user)	Decides to adopt the system. Cares about cost, patient satisfaction and fewer no-shows.	Reads reports and results through the admin.
Maintainers and examiners	Indirect	Read documentation, code and tests.	Assess, maintain and extend the product.

Personas (fictional)

Personas are made-up people that stand for each user class. They help the team design for real needs. They should be refined after the stakeholder interviews listed in Appendix C.

Persona	Background	Goals	Frustrations today
Rahima Akter, 34, school teacher (Patient)	Uses a smartphone daily. Books care for herself and her mother. Works fixed hours.	Book quickly in the evening; know when to leave home; manage her mother's visits.	Phone lines busy; hours lost in the waiting room; cannot tell when the doctor is running late.
Karim Uddin, 62, retired (Patient)	Not comfortable with apps. Often relies on his son or on reception.	Be seen without confusion; large clear text; ask a person for help.	Does not understand serial numbers; hard to hear his name called.
Shirin Sultana, 28, receptionist	Handles phone calls, walk-ins and arguments from the front desk.	Register people fast; avoid double bookings; show a fair queue.	Paper register errors; patients say "I was here first".
Dr. Imran Hossain, 45, general physician	Sees about 30 patients in an evening session.	See who is next and who has waited longest; keep notes short; control his own schedule.	Unpredictable patient flow; chasing history and notes.
Farhan Ahmed, 40, clinic manager	Runs daily operations and staff.	Reduce no-shows; understand doctor utilisation; control access.	No data on waiting times or no-shows; staff mistakes.

2.2 User Needs

User	Needs (what they need to be true)	Stories
Patient	• Book, change or cancel without calling the clinic. • Know my position and how long I will wait, and be told when my turn is near. • Be told quickly if the doctor cancels. • Use my phone, in my language, and keep my data private and under my control (change my number, copy or delete my data). • Notice when I am called, by sight or by sound. • Manage my family's appointments from one account.	AUTH, BKG, QUE-04/05/11/13/15, NOT, SYS
Receptionist	• Serve phone callers and walk-ins quickly. • Never double-book a slot. • Handle emergencies and exceptions with a clear record of why. • Keep working if the system is down. • Help a patient who lost their phone or changed their number.	AUTH-09, BKG-09/10, QUE-01/03/09/10/14, SYS-04
Doctor	• Control my own availability and leave. • See an ordered live queue and call the next patient with one click. • Record notes and prescriptions quickly and see relevant history.	SCH-02/03, QUE-06/07/08/12, VIS
Clinic admin	• Control who has access and see who did what. • Set clinic rules without developers. • Understand no-shows and waiting times.	AUTH-05/07, SCH-01/04/05, ADM, SYS-03
Clinic owner (buyer)	• Fewer no-shows and idle doctor time. • Happier patients; low running cost. • Safe handling of patient data.	NOT-02, ADM, SEC, PRIV

2.3 Operating Environment

Component	Requirement
Users' devices	Smartphones, tablets, laptops and PCs with a current web browser (see COMP-01). No app installation needed.
Server	A web server and application server on a cloud or local machine. The technology stack is chosen in the SDS; it must support the architecture pattern and the targets in Section 3.2.
Database	A relational database is recommended so that uniqueness rules (no double booking) can be enforced by the database itself (REL-02).
Network	Internet access at the clinic and for patients. The clinic PC should have a reliable connection; a printed schedule is the fallback (SYS-04).
External services	An SMS gateway (sandbox or simulated during development), an email service, and optionally a QR scanner (phone camera).
Waiting-room display	Any 1080p screen or TV running a browser in full-screen mode.
Time and locale	Server time synchronised; time zone Asia/Dhaka (UTC+6); Bangla and English text.

2.4 Constraints

ID	Type	Constraint
C-01	**Schedule**	The project runs during the semester (project work from week 3 to week 11) and ends with a demonstration, report and viva.
C-02	**Team**	Built by a small student team 4 members. Work is split so that every member contributes code, tests and documents.

ID	Type	Constraint
C-03	**Budget**	No budget. Use free and open-source tools and free tiers. SMS is simulated or sandboxed; email or in-app messages are the real fallback.
C-04	**Architecture**	The code shall follow one recognised pattern (MVC, ECB or CQRS) and be explained in the SDS.
C-05	**Process and tools**	Use Git and GitHub (with pull-request code review), GitHub Wiki, Trello, Discord and Toggl as required by the project assessment.
C-06	**Quality**	Source code must be documented and follow a coding standard; unit tests are required; the UI is judged with Nielsen's heuristics.
C-07	**Documents**	The SRS and SDS follow the course format (SDS contains a use case diagram, a DFD and a state diagram).
C-08	**Data and legal**	Only synthetic data may be used. Personal-data handling must follow the principles in PRIV and LEGAL-01.
C-09	**Language**	The user interface must support Bangla and English.

2.5 Assumptions and Dependencies

If an assumption turns out to be false, the requirements affected may have to change. The team should confirm these with stakeholders (Appendix C).

ID	Assumption or dependency	If it is false…
A-01	One clinic site with several doctors; each doctor has one consulting room.	Multi-branch support and room allocation would be needed.
A-02	A doctor sees one patient at a time.	BR-11 and the queue model change.
A-03	Most patients have a mobile phone and internet; others use reception.	More reception features and printed material are needed.
A-04	An SMS service (or a free sandbox) is available; email is a fallback.	More reliance on in-app messages and email.
A-05	The clinic works fixed daily sessions and publishes them as weekly schedules.	Flexible or on-call scheduling would need new rules.
A-06	An average consultation takes 10 to 20 minutes.	Default slot length and EWT defaults need adjusting.
A-07	The reception desk has a PC with internet access.	A mobile-first staff interface would be needed.
A-08	No billing or payment is handled in this release.	Payment and invoice requirements would be added.
A-09	At least three stakeholders (for example a receptionist, a doctor, a patient) are available to review requirements.	More assumptions stay unvalidated; risk of building the wrong thing rises (Lecture 1: requirements mismatch).
A-10	A mobile number identifies one patient account; dependents share the guardian's number.	Account and OTP design change.
A-11	Server clocks are accurate.	Slot, grace-period and reminder timing would be unreliable.
A-12	Reception can check a patient's identity in person (for example, with a national ID card or earlier visit details) before changing a registered mobile number.	Account takeover through number changes becomes possible; AUTH-09 would need a stronger check.

3. Requirements

> **In plain words:** Section 3.1 lists what users can do, as user stories. Section 3.2 says how well the system must perform. Sections 3.3 to 3.5 give the rules, interfaces and data behind them.

3.1 Functional Requirements (User Stories)

The functional requirements are written as user stories **with confirmation criteria**, as required by the course SRS guide. There are **62 stories** in **8 epics**, worth **234 story points** in total.

Epic	Stories	Must	Should	Could	Points
AUTH — Accounts and Access	9	4	5	0	31
SCH — Doctors and Schedules	6	3	2	1	20
BKG — Appointment Booking	11	6	3	2	43
QUE — Live Queue	15	8	5	2	66
VIS — Consultations and Records	7	1	5	1	30
NOT — Notifications	5	2	2	1	14
ADM — Reports and Oversight	4	0	3	1	15
SYS — Cross-cutting Needs	5	2	3	0	15
Total	**62**	**26**	**28**	**8**	**234**

How to read a story: the left cell is the story, the middle cell is how we will check it is done, and the right cell shows priority, points and the related rules, risks and non-functional requirements.

3.1.1 Epic AUTH — Accounts and Access

Who can enter the system, and what each person is allowed to do.

ID	User story	Confirmation (how we know it is done)	Priority · Links
AUTH-01	**As a** new patient, **I want** to register with my mobile number and verify it with a one-time code (OTP), **so that** I can book appointments without visiting the clinic.	**1. Given** a valid Bangladeshi mobile number (+8801XXXXXXXXX), **when** I enter the correct OTP within 5 minutes, **then** my account is created and I am logged in. **2. Given** I enter a wrong OTP 3 times, **when** I try again, **then** registration is blocked for 15 minutes and I see a clear message.	**MUST** **5 pts** Links: SEC-04, USAB-02
AUTH-02	**As a** registered patient, **I want** to log in with my mobile number and an OTP, and log out, **so that** only I can see and manage my appointments.	**1. Given** I am registered, **when** I enter the correct OTP, **then** I reach my dashboard. **2. Given** I have been inactive for 30 minutes, **when** I click anything, **then** I must log in again.	**MUST** **3 pts** Links: SEC-05
AUTH-03	**As a** patient, **I want** to keep my profile up to date (name, age, gender, address, emergency contact), **so that** the clinic always has correct details.	**Given** I change my emergency contact, **when** I save, **then** the new value shows immediately and the change is written to the audit log.	**SHOULD** **2 pts** Links: SEC-06

ID	User story	Confirmation (how we know it is done)	Priority · Links
AUTH-04	**As a** patient, **I want** to add family members (dependents) and book on their behalf, **so that** I can manage my child's or parent's care from one account.	**1. Given** I added my mother with her name and date of birth, **when** I start a booking, **then** I can choose her as the patient, and her name appears on the appointment and in the queue. **2. Given** a dependent has visits, **when** another user tries to view them, **then** access is refused; only the guardian account and the treating doctor can see them.	SHOULD **5 pts** Links: PRIV-05
AUTH-05	**As a** clinic admin, **I want** to create staff accounts (doctor, receptionist), assign roles and deactivate accounts, **so that** only authorised people use the system.	**1. Given** I create a receptionist account, **when** they log in for the first time, **then** they must set a new password. **2. Given** a staff account is deactivated, **when** that person tries to log in, **then** access is denied but their past records remain.	MUST **5 pts** Links: SEC-01, SEC-02
AUTH-06	**As a** staff member, **I want** to log in with email and password and reset a forgotten password, **so that** I can work securely without waiting for IT help.	**1. Given** I enter a wrong password 5 times, **when** I try again, **then** my account is locked for 15 minutes. **2. Given** I request a password reset, **when** I use the emailed link within 30 minutes, **then** I can set a new password and the link stops working.	MUST **3 pts** Links: SEC-02, SEC-04
AUTH-07	**As a** clinic admin, **I want** to see an audit log of sensitive actions (logins, record views, cancellations, role changes), **so that** misuse can be traced and explained.	**1. Given** a doctor opens a patient's visit notes, **when** I view the audit log, **then** I see who did it, what they did and when. **2. Given** I open the audit log, **when** I look for edit or delete options, **then** there are none.	SHOULD **3 pts** Links: SEC-06
AUTH-08	**As a** patient, **I want** to change my registered mobile number by confirming a one-time code sent to the new number, **so that** I keep my account and appointments when I change my number.	**1. Given** I am logged in and enter a new mobile number, **when** I enter the correct OTP sent to that number within 5 minutes, **then** my account uses the new number, I stay logged in, a notice is sent to my old number and the change is written to the audit log. **2. Given** the new number already belongs to another account, **when** I try to use it, **then** the change is refused with a clear message and nothing changes.	SHOULD **3 pts** Links: SEC-04, SEC-06, A-10
AUTH-09	**As a** receptionist, **I want** to update a patient's mobile number after checking their identity, **so that** a patient who lost their phone can still reach their account and appointments.	**1. Given** a patient cannot receive codes on the old number, **when** I enter the new number and confirm that I checked their identity, **then** the number is updated, the patient's active sessions end and the change is logged with my name. **2. Given** I am a patient or a doctor, **when** I try to open this function, **then** access is refused.	SHOULD **2 pts** Links: A-12, SEC-01, SEC-06, R-04

3.1.2 Epic SCH — Doctors and Schedules

How doctors and the clinic define the times when appointments can be booked.

ID	User story	Confirmation (how we know it is done)	Priority · Links
SCH-01	**As a** clinic admin, **I want** to add and edit doctor profiles (name, specialty, room, photo, default slot length), **so that** patients can find the right doctor and the system can build schedules.	**1. Given** I fill in all required fields, **when** I save, **then** the doctor appears in patient search with specialty and room. **2. Given** a doctor has future appointments, **when** I try to delete the doctor, **then** the system offers to deactivate instead.	MUST **3 pts**
SCH-02	**As a** doctor, **I want** to set my weekly availability (days, hours, breaks, slot length), **so that** patients can only book times when I am really available.	**1. Given** Monday 17:00–21:00 with 15-minute slots, **when** I save, **then** 16 slots are offered for every Monday. **2. Given** I add a break from 19:00 to 19:30, **when** I save, **then** those 2 slots disappear from the patient view.	MUST **5 pts** Links: BR-01
SCH-03	**As a** doctor, **I want** to mark days or hours when I am on leave, **so that** no new bookings are made and affected patients are looked after.	**Given** I mark 20–22 Oct as leave, **when** I save, **then** no slots are offered for those days, existing appointments become “Needs Rescheduling” and the affected patients are notified (NOT-03).	MUST **5 pts** Links: BR-13, NOT-03, R-07
SCH-04	**As a** clinic admin, **I want** to set clinic-wide rules (booking window, cancellation cut-off, grace period, daily limits), **so that** clinic policy can change without changing code.	**Given** I change the cancellation cut-off from 2 to 4 hours, **when** a patient tries to cancel 3 hours before the slot, **then** the system blocks the cancellation.	SHOULD **3 pts** Links: MAIN-05
SCH-05	**As a** clinic admin, **I want** to record clinic closed days and public holidays, **so that** nobody books on a day we are closed.	**Given** a day is marked closed, **when** a patient looks for slots, **then** none are shown and a “Clinic closed” message appears.	SHOULD **2 pts**
SCH-06	**As a** receptionist, **I want** to hold back or release individual slots (for example, to reserve time for emergencies), **so that** the clinic keeps flexibility for urgent cases.	**Given** I block the 18:00 slot, **when** a patient searches, **then** that slot is not shown; when I release it, it appears again.	COULD **2 pts**

3.1.3 Epic BKG — Appointment Booking

Finding a doctor, choosing a time, and changing or cancelling it.

ID	User story	Confirmation (how we know it is done)	Priority · Links
BKG-01	**As a** patient, **I want** to search for doctors by name or specialty, **so that** I can quickly find who can treat me.	**1. Given** I type “cardio”, **when** I search, **then** doctors whose name or specialty matches are listed. **2. Given** nothing matches, **when** I search, **then** I see a friendly message and the list of all specialties.	MUST **3 pts**
BKG-02	**As a** patient, **I want** to see a doctor’s free time slots for the coming days, **so that** I can choose a time that suits me.	**Given** a doctor has availability, **when** I open their page, **then** I see free slots within the booking window (Dhaka time); taken, blocked and past slots cannot be selected.	MUST **3 pts** Links: BR-03
BKG-03	**As a** patient, **I want** to book a free slot and add an optional reason for my visit, **so that** I have a confirmed time and a reference number.	**1. Given** a free slot, **when** I confirm, **then** an appointment with status “Booked” and a reference number is created and shown. **2. Given** two patients choose the same slot at the same moment, **when** both confirm, **then** exactly one succeeds and the other sees “Slot just taken” with nearby alternatives.	MUST **8 pts** Links: BR-01, BR-03, REL-02, R-01

ID	User story	Confirmation (how we know it is done)	Priority · Links
BKG-04	**As a** patient, **I want** to see my upcoming and past appointments with their status, **so that** I always know where I stand.	**Given** I have 2 upcoming and 5 past appointments, **when** I open “My appointments”, **then** upcoming ones are first, and each shows doctor, time and status.	MUST **3 pts**
BKG-05	**As a** patient, **I want** to cancel an appointment before the cut-off time, **so that** the slot can be given to someone else.	**1. Given** my appointment is more than 2 hours away, **when** I cancel and confirm, **then** the status becomes “Cancelled”, the slot is free and I get a confirmation. **2. Given** my appointment is less than 2 hours away, **when** I try to cancel, **then** the system blocks it and tells me to call reception.	MUST **3 pts** Links: BR-04
BKG-06	**As a** patient, **I want** to reschedule an appointment to another free slot in one step, **so that** I do not lose my place by cancelling first.	**1. Given** a booked appointment before the cut-off, **when** I choose another free slot, **then** the new slot is booked and the old slot freed together. **2. Given** the new slot was just taken, **when** I confirm, **then** my original appointment stays unchanged and I am told why.	SHOULD **5 pts** Links: BR-04
BKG-07	**As a** patient, **I want** to be stopped if a booking clashes with another of my appointments or I hold too many, **so that** I do not accidentally double-book myself or hoard slots.	**1. Given** I have an active appointment 17:00–17:15, **when** I try to book another doctor at 17:00, **then** the system blocks it and explains why. **2. Given** I already have 3 upcoming active appointments, **when** I try to book a 4th, **then** the system blocks it and explains the limit.	SHOULD **3 pts** Links: BR-02, BR-16, R-11
BKG-08	**As a** patient, **I want** to join a waiting list for a fully booked day and be told when a slot opens, **so that** I have a chance to get an appointment after a cancellation.	**Given** all slots are taken and I joined the waiting list, **when** someone cancels, **then** the first person on the list is notified and has 15 minutes to claim the slot.	COULD **5 pts**
BKG-09	**As a** receptionist, **I want** to book an appointment by phone for a caller (finding their record or creating a new one), **so that** people without smartphones can still get appointments.	**1. Given** a caller’s mobile number, **when** I search, **then** the existing patient is found, or I can create a record with only name, phone and age. **2. Given** I select a free slot, **when** I confirm, **then** an appointment of type “Reception” is created.	MUST **5 pts** Links: USAB-03
BKG-10	**As a** receptionist, **I want** to cancel or reschedule any appointment and record the reason (for example, the doctor is delayed), **so that** late changes are handled safely.	**Given** a booked appointment, **when** I cancel it with the reason “Doctor unavailable”, **then** the status becomes “Cancelled”, the reason is stored and the patient is notified.	SHOULD **3 pts** Links: NOT-01
BKG-11	**As a** patient, **I want** to get an appointment slip with my reference number and a QR code, **so that** I can show it at the clinic for fast check-in.	**Given** a booked appointment, **when** I open it, **then** I can view, download or print the slip.	COULD **2 pts**

3.1.4 Epic QUE — Live Queue

What happens on the day: check-in, who is next, and how long people wait.

ID	User story	Confirmation (how we know it is done)	Priority · Links
QUE-01	**As a** receptionist, **I want** to check in a patient with an appointment when they arrive, **so that** they join the queue in the right place.	**1. Given** an appointment today and arrival within the check-in window, **when** I check in the patient, **then** the status becomes "Checked-In", a token is issued and the patient enters the queue. **2. Given** the patient arrives after the grace period but before the no-show limit, **when** I check in the patient, **then** they are checked in as a late arrival: they join the queue from their check-in time, the appointment is not counted as a no-show and I am told why. **3. Given** the patient arrives after the no-show limit, **when** I try to check in the appointment, **then** I am offered to register them as a walk-in (QUE-03).	MUST 5 pts Links: BR-05, BR-08, BR-14
QUE-02	**As a** patient, **I want** to check myself in by scanning my QR slip when I arrive, **so that** reception staff are free for other work.	**1. Given** a valid QR slip within the check-in window, **when** I scan it at the clinic screen, **then** I am checked in and I see my token. **2. Given** a QR slip for a different day, **when** I scan it, **then** it is rejected with a clear message.	COULD 5 pts Links: BR-05
QUE-03	**As a** receptionist, **I want** to register a walk-in patient and add them to a doctor's queue in under 30 seconds, **so that** unplanned visitors are served fairly.	**Given** a walk-in with name, phone and age, **when** I choose a doctor, **then** the patient is created (or found) and gets the next token in that doctor's queue.	MUST 5 pts Links: BR-08, USAB-03
QUE-04	**As a** patient, **I want** to see my token, queue position and estimated waiting time live, **so that** I can decide whether to wait or step out.	**1. Given** I am checked in and 3rd in line, **when** someone ahead is called, **then** my position changes to 2nd within 5 seconds and the estimate updates. **2. Given** an estimate is shown, **when** I read it, **then** it is a range (for example "about 20–30 min"), not an exact time.	MUST 8 pts Links: BR-12, PERF-03, R-08
QUE-05	**As a** patient (before leaving home), **I want** to see how many people are waiting for my doctor today and whether the doctor is running late, **so that** I can time my arrival.	**Given** a doctor has 6 people waiting, **when** I open the doctor's page, **then** I see "6 waiting, about X min" with the time of the last update and no personal names.	SHOULD 3 pts Links: PRIV-03
QUE-06	**As a** doctor, **I want** to see my live queue with order, patient name, age, reason and waiting time, **so that** I know who is next and who has waited longest.	**Given** patients are waiting, **when** I open my queue, **then** they are ordered by the queue rule, emergencies are highlighted and the list updates without refreshing.	MUST 5 pts Links: BR-08
QUE-07	**As a** doctor, **I want** to call the next patient with one click, **so that** the queue moves forward and patients can see it.	**1. Given** the queue is not empty and nobody is in consultation, **when** I click "Call next", **then** the first patient becomes "In Consultation" and the patient's live queue screen (QUE-04) and my queue show it within 5 seconds. **2. Given** a patient is already in consultation, **when** I click "Call next", **then** the system asks me to complete the current one first.	MUST 5 pts Links: BR-11, PERF-03

ID	User story	Confirmation (how we know it is done)	Priority · Links
QUE-08	**As a** doctor, **I want** to finish a consultation, or skip a called patient who does not respond, **so that** the queue stays accurate.	**1. Given** a patient is in consultation, **when** I click “Complete”, **then** the status becomes “Completed” and the duration is recorded. **2. Given** a called patient does not respond, **when** I click “Skip”, **then** they return to waiting behind current waiters; after 2 skips I can no longer skip them and reception decides (QUE-10).	MUST 5 pts Links: BR-10
QUE-09	**As a** receptionist, **I want** to flag a patient as an emergency, with a reason, **so that** urgent cases are never stuck behind routine visits and move to the front of the queue.	**Given** a patient is waiting, **when** I flag “Emergency” and enter a reason, **then** the patient becomes first in the queue, the doctor is alerted and the action is logged with my name.	MUST 3 pts Links: BR-09, SAFE-02, R-03
QUE-10	**As a** receptionist, **I want** to move or remove a patient in the queue and record the reason, **so that** I can fix exceptions such as a patient who left.	**1. Given** a patient is waiting, **when** I remove them with the reason “Left clinic”, **then** the entry is removed, the others’ positions update and the reason is stored. **2. Given** a patient has been skipped twice, **when** I open the queue, **then** I see them marked “Reception to decide” and I can return them to the queue or remove them.	MUST 3 pts Links: BR-10
QUE-11	**As a** patient in the waiting room, **I want** to see a public screen showing “Now serving” token numbers, **so that** I know when my turn is without asking.	**Given** the display screen is open, **when** a doctor calls the next patient, **then** the screen shows doctor, token and room within 5 seconds, with no names or medical information.	SHOULD 5 pts Links: PRIV-03, COMP-03
QUE-12	**As a** doctor, **I want** to pause the queue for a break or urgent matter and then resume it, **so that** waiting patients are informed instead of left guessing.	**Given** I click “Pause” with an optional reason, **when** patients look at their screen, **then** they see “Doctor on a short break” and estimates include the pause; “Resume” restores normal flow.	SHOULD 3 pts
QUE-13	**As a** patient, **I want** to get a waiting-time estimate that reflects how fast my doctor really works, **so that** the waiting time I see is trustworthy.	**Given** a doctor’s last 20 consultations averaged 12 minutes, **when** I am 3rd in line, **then** the estimate uses about 12 minutes for each patient ahead of me.	SHOULD 5 pts Links: BR-12, R-08
QUE-14	**As a** receptionist, **I want** to have patients who do not arrive in time marked “No-Show” automatically, **so that** the schedule and records stay accurate without manual work.	**1. Given** a booked appointment is not checked in by the no-show limit (slot start + 60 minutes), **when** the time passes, **then** the status becomes “No-Show” and the doctor’s queue is not affected. **2. Given** a patient marked “No-Show” arrives later the same day, **when** I reverse the No-Show and enter a reason, **then** the appointment becomes “Checked-In” from the check-in time and the reversal is logged with my name.	SHOULD 3 pts Links: BR-06
QUE-15	**As a** patient (or a person who cannot see or hear well), **I want** the waiting-room display to show and also announce my token when I am called, **so that** I do not miss my turn.	**1. Given** announcements are switched on, **when** the doctor calls my token, **then** the display shows token and room in large text and plays a chime followed by a spoken announcement in the selected language. **2. Given** the display has no sound, **when** a token is called, **then** the display still shows it clearly, so nobody depends on sound alone.	COULD 3 pts Links: USAB-07, QUE-11, PRIV-03

3.1.5 Epic VIS — Consultations and Records

What the doctor records during and after the visit.

ID	User story	Confirmation (how we know it is done)	Priority · Links
VIS-01	**As a** doctor, **I want** to record visit notes (complaint, findings, diagnosis, advice) for the patient I am seeing, **so that** the patient's history is complete.	**Given** a patient is in consultation, **when** I save my notes, **then** they are stored against that visit with the time and my name.	MUST **5 pts** Links: BR-15
VIS-02	**As a** doctor, **I want** to view the history of a patient who is in my queue or has an appointment with me, **so that** I treat them with full context.	**1. Given** a patient is in my queue, **when** I open their history, **then** I see their earlier visits at this clinic. **2. Given** a patient is not in my queue or appointments, **when** I try to open their history, **then** access is refused and the attempt is logged.	SHOULD **3 pts** Links: BR-15, SEC-01, PRIV-05
VIS-03	**As a** doctor, **I want** to write a prescription as a list of medicines (name, dose, how often, how long) and print or download it, **so that** the patient leaves with a clear, readable prescription.	**Given** a visit is in progress, **when** I add 2 medicines and click "Print", **then** a prescription is produced with clinic, doctor, date, patient and medicines.	SHOULD **8 pts**
VIS-04	**As a** doctor, **I want** to suggest a follow-up date that creates a ready-to-confirm booking for the patient, **so that** follow-ups are not forgotten.	**Given** I choose "follow-up in 7 days", **when** I save, **then** the patient is notified with a suggested slot to confirm or change.	SHOULD **5 pts**
VIS-05	**As a** patient, **I want** to read my visit summaries (advice and prescription), **so that** I keep my own health records.	**Given** I have a completed visit, **when** I open "My records", **then** I see the date, doctor, advice and prescription shared with me.	SHOULD **3 pts** Links: PRIV-05
VIS-06	**As a** receptionist, **I want** to record basic vitals (weight, blood pressure, temperature) at check-in, **so that** the doctor starts with this information.	**Given** a patient is checked in, **when** I enter their vitals, **then** the doctor sees them with the patient in the queue.	COULD **3 pts**
VIS-07	**As a** doctor, **I want** to correct an old note through a visible amendment instead of a silent edit, **so that** the record stays trustworthy.	**Given** a note is older than 24 hours, **when** I change it, **then** the original is kept and the amendment is stored with my name and the time.	SHOULD **3 pts** Links: BR-15

3.1.6 Epic NOT — Notifications

Messages that keep patients informed without calling the clinic.

ID	User story	Confirmation (how we know it is done)	Priority · Links
NOT-01	**As a** patient, **I want** to get a confirmation when an appointment is booked, cancelled or rescheduled, **so that** I am sure it was recorded.	**Given** I book an appointment, **when** it is saved, **then** I receive an SMS (or email, or in-app message) within 1 minute with doctor, date, time and reference number.	MUST **3 pts** Links: REL-04, R-06
NOT-02	**As a** patient, **I want** to get a reminder 24 hours and 2 hours before my appointment, **so that** I do not forget, and the clinic has fewer no-shows.	**Given** my appointment is in 24 hours, **when** the time arrives, **then** I receive a reminder with a link to cancel or reschedule.	SHOULD **3 pts**

ID	User story	Confirmation (how we know it is done)	Priority · Links
NOT-03	**As a** patient, **I want** to be told promptly if my doctor goes on leave and my appointment is affected, **so that** I can rebook without a wasted trip.	**Given** a doctor's leave affects my appointment, **when** the leave is saved, **then** I am notified within 1 minute with a link to my appointment, where I can cancel it and book a new slot.	MUST **3 pts** Links: SCH-03, BR-13, R-07
NOT-04	**As a** patient, **I want** to get an alert when I am two patients away, next, or called, **so that** I can be at the door at the right time.	**1. Given** I am two patients away, **when** the queue advances, **then** I receive an alert. **2. Given** the doctor calls my token, **when** the call is made, **then** I receive an alert with my token and the room.	SHOULD **3 pts** Links: QUE-07
NOT-05	**As a** patient, **I want** to choose how I am notified (SMS, email, in-app) and in which language, **so that** messages reach me the way I prefer.	**Given** I choose email and Bangla, **when** a message is sent, **then** it arrives by email in Bangla.	COULD **2 pts**

3.1.7 Epic ADM — Reports and Oversight

Helping the clinic manager see how the clinic is running.

ID	User story	Confirmation (how we know it is done)	Priority · Links
ADM-01	**As a** clinic admin, **I want** to see a daily dashboard (appointments, walk-ins, no-shows, average wait per doctor), **so that** I can see how the clinic is running today.	**Given** a clinic day is in progress, **when** I open the dashboard, **then** I see live counts and average waits that match the underlying records.	SHOULD **5 pts**
ADM-02	**As a** clinic admin, **I want** to produce an appointment report by date range, doctor and status, and export it to CSV, **so that** I can analyse and share the results.	**Given** a date range containing 100 appointments, **when** I export, **then** a CSV file with the same 100 rows opens correctly in a spreadsheet.	SHOULD **5 pts**
ADM-03	**As a** clinic admin, **I want** to view average waiting time and consultation length per doctor over time, **so that** I can improve schedules and slot lengths.	**Given** a month of data, **when** I open the analytics page, **then** I see a chart per doctor of average wait and average consultation time.	SHOULD **3 pts**
ADM-04	**As a** clinic admin, **I want** to see which patients are restricted for repeated no-shows and lift a restriction, **so that** patients are treated fairly.	**1. Given** a patient has 3 no-shows within 60 days, **when** they try to book online, **then** they are told to book through reception for the next 14 days. **2. Given** a patient is restricted, **when** I lift the restriction, **then** they can book online again immediately and the action is logged.	COULD **2 pts** Links: BR-07, R-11

3.1.8 Epic SYS — Cross-cutting Needs

Needs that touch the whole system: language, phone use, backups, and privacy.

ID	User story	Confirmation (how we know it is done)	Priority · Links
SYS-01	**As a** patient, **I want** to use the system in Bangla or English and switch at any time, **so that** I can understand every screen.	**Given** I switch to Bangla, **when** I move through the system, **then** every label, message and notification appears in Bangla (coverage rules: USAB-05).	SHOULD **3 pts** Links: USAB-05

ID	User story	Confirmation (how we know it is done)	Priority · Links
SYS-02	**As a** patient, **I want** to use the whole system comfortably on my phone, **so that** I do not need a computer.	**Given** a phone screen at the minimum width in COMP-02, **when** I register and book, **then** nothing needs sideways scrolling and every button is easy to tap.	MUST **3 pts** Links: COMP-02
SYS-03	**As a** clinic admin, **I want** to have the system back up its data automatically every day, **so that** a failure does not lose appointments or records.	**Given** a daily backup has run, **when** the backup is restored into a test environment, **then** the data up to the last backup is recovered (targets: REL-03).	SHOULD **2 pts** Links: REL-03
SYS-04	**As a** receptionist, **I want** to print today's schedule for each doctor, **so that** the clinic can keep working if the system or internet is down.	**Given** a day with 20 appointments, **when** I click "Print", **then** I get a one-page list per doctor with time, name, phone and status.	MUST **2 pts** Links: R-09
SYS-05	**As a** patient, **I want** to request a copy of my data or the deletion of my account, **so that** I stay in control of my personal information.	**1. Given** I request a copy of my data, **when** the request is processed, **then** I can download my profile, appointments and visit summaries as a readable file. **2. Given** I request deletion, **when** the admin approves, **then** my personal details are anonymised and records the clinic must keep are retained under policy.	SHOULD **5 pts** Links: PRIV-02, SEC-06

3.2 Non-Functional Requirements

Non-functional requirements say how well the system must work. They are grouped by the ISO/IEC 25010 quality model and each has a measurable target. Verification: T = test, I = inspection, D = demonstration, A = analysis.

3.2.1 Performance Efficiency

ID	Requirement	Verify
PERF-01	The system shall load any main page (search, booking, queue, dashboard) in 2 seconds or less for 95% of requests on a 4G connection (5 Mbps or more) under normal load.	T
PERF-02	Confirming a booking shall complete in 3 seconds or less for 95% of attempts.	T
PERF-03	A queue change (check-in, call next, skip, emergency flag) shall appear on patient, doctor and display screens within 5 seconds, including the time the screens need to refresh.	T, D
PERF-04	The system shall support at least 10 doctors, 200 concurrent users and 1,000 appointments per day while still meeting PERF-01 to PERF-03.	T
PERF-05	The waiting-time estimate shown at check-in shall be within $\pm$10 minutes of the actual waiting time (measured from the middle of the displayed range) in at least 90% of cases, in simulated queue runs of at least 200 patients with consultation times of 10 to 20 minutes (A-06).	T, A

3.2.2 Reliability

ID	Requirement	Verify
REL-01	During clinic hours the system should be available at least 99.5% of the time, measured monthly (about 4 minutes of downtime per 14-hour day). For the course demo, the system shall run without a crash during the demonstration.	A, D
REL-02	The system shall never create two active appointments for the same doctor and slot. This shall be enforced by a database constraint and not only by application code. Test: 100 simultaneous booking attempts for one slot produce exactly one success.	T
REL-03	All data shall be backed up automatically every 24 hours and be restorable with at most 24 hours of data loss (RPO) and within 4 hours (RTO).	T, I
REL-04	If the SMS or email service fails, the main action (for example booking) shall still succeed. The system shall retry sending up to 3 times and show the message in the user's in-app inbox.	T

3.2.3 Security

ID	Requirement	Verify
SEC-01	The system shall check the user's role on the server for every request. A patient shall never reach staff functions, and staff shall reach only the functions of their own role (role $\times$ function test matrix).	T
SEC-02	Passwords shall have at least 8 characters and be stored only as salted hashes (bcrypt or Argon2), never in plain text or in logs.	I, T
SEC-03	All communication shall use HTTPS (TLS 1.2 or higher). Session cookies shall be marked Secure and HttpOnly.	I, T
SEC-04	OTP codes shall expire after 5 minutes and allow at most 3 attempts. After 5 failed logins, the account or number shall be locked for 15 minutes.	T
SEC-05	Sessions shall expire after 30 minutes of inactivity for patients and 15 minutes for staff.	T
SEC-06	The system shall record who did what and when for sensitive actions (login, viewing or editing visit notes, cancellation, role changes, emergency flag, changing a patient's mobile number, data export and deletion). Audit entries shall not be editable or deletable in the application and shall be kept for at least 1 year.	T, I
SEC-07	The system shall validate all input and use parameterised queries to defend against the OWASP Top 10 (injection, cross-site scripting, request forgery and others). A scan with OWASP ZAP shall show no High findings before the final demo.	T

3.2.4 Privacy and Data Protection

ID	Requirement	Verify
PRIV-01	The system shall collect only the personal data needed for appointments and care (data minimisation).	I
PRIV-02	A patient shall be able to see and export their own data and request deletion or anonymisation (SYS-05). Records the clinic must keep by law or policy are retained.	D, T
PRIV-03	Public screens (waiting-room display and "people waiting" counts) shall show only token numbers and counts, never names, phone numbers or reasons for visit.	I, D
PRIV-04	Development, testing and demonstration shall use synthetic (made-up) data only.	I
PRIV-05	Clinical notes shall be readable only by the authoring doctor and by any doctor who currently has that patient in their queue or appointment list. The patient sees only the summary (advice and prescription). Receptionists and admins shall not see clinical notes.	T

3.2.5 Usability, Accessibility and Localisation

ID	Requirement	Verify
USAB-01	The interface shall follow Nielsen's 10 usability heuristics. A heuristic evaluation by at least 3 team members shall find no unresolved severity 3 or 4 problems.	I, D
USAB-02	A first-time patient shall be able to register and book an appointment in 5 minutes or less and 8 steps or fewer without help, tested with at least 5 people.	T
USAB-03	A receptionist shall be able to register a walk-in and add them to a queue in 30 seconds or less and 3 screens or fewer.	T
USAB-04	The main patient and staff flows shall meet WCAG 2.1 Level AA (for example text contrast of at least 4.5:1, full keyboard use, labelled form fields, text resizable to 200%).	I, T
USAB-05	All screens, messages and notifications shall be available in Bangla and English, switchable at any time. Dates and times shall be shown in Asia/Dhaka (UTC+6) and mobile numbers in +880 format.	I, D
USAB-06	Error messages shall say what went wrong and what to do next, in plain language and without technical codes.	I
USAB-07	Every patient call in the waiting room shall be shown on screen and, when the clinic switches announcements on, also given by sound, so that nobody depends on sight alone or on sound alone. Calls use token numbers, not names (PRIV-03).	D

3.2.6 Compatibility and Portability

ID	Requirement	Verify
COMP-01	The system shall work on the latest two major versions of Chrome, Edge, Firefox and Safari, and on mobile Chrome (Android) and Safari (iOS).	T
COMP-02	The interface shall be responsive from 360 px to 1920 px width, with touch targets of at least 44 $\times$ 44 px on mobile.	T, I
COMP-03	The waiting-room display shall run in a browser at 1920 $\times$ 1080 in full-screen mode, with text readable from 5 metres.	D
COMP-04	The system shall be deployable from written steps in 30 minutes or less on a clean machine.	D

3.2.7 Maintainability

ID	Requirement	Verify
MAIN-01	The code shall use a layered architecture following one chosen pattern (MVC, ECB or CQRS), keeping business rules separate from screens and database code.	I
MAIN-02	Automated unit tests shall cover at least 70% of the business-logic code, and every business rule that is built in the delivered release (Appendix A.2) shall have at least one test.	T
MAIN-03	Code shall follow a written coding standard checked by a linter. Every public class and function shall have a documentation comment (coverage of at least 80%).	I
MAIN-04	Every change shall enter the main branch through a pull request with at least one peer review and passing automated tests (CI).	I
MAIN-05	Clinic rules (booking window, cut-off, grace period, limits) shall be settings that can change without editing code.	T

3.2.8 Safety and Compliance

ID	Requirement	Verify
SAFE-01	The system shall not give diagnosis or medical advice. Booking pages shall show: “If this is an emergency, call 999 or go to the nearest hospital.”	I
SAFE-02	An emergency flag shall always override the normal queue order and shall be logged with who set it, when and why.	T
LEGAL-01	The system should follow applicable Bangladeshi law on health and personal data. GDPR principles (lawfulness, minimisation, purpose limitation, accuracy, security, access) are used as an international reference. A legal review is needed before any real deployment.	I

3.3 Business Rules

Business rules are the “real rules” that the system must enforce. Defaults can be changed by the admin (SCH-04). Each rule is testable. Every rule that is built has a unit test (MAIN-02); a rule whose story is not built in a release is tested when it is built (Appendix A.2).

ID	Rule	Default	Stories
BR-01	**Slot uniqueness.** A slot can hold at most one active appointment (Booked, Checked-In or In Consultation).	Slot 15 min	SCH-02, BKG-03
BR-02	**No clashes.** A person cannot hold two active appointments that overlap in time, with any doctor.	—	BKG-07
BR-03	**Booking window.** Patients may book from 14 days ahead up to 30 minutes before the slot starts. Reception may book until the slot ends.	14 days / 30 min	BKG-02, BKG-03
BR-04	**Cancel and reschedule cut-off.** Patients may cancel or reschedule up to 2 hours before the slot. After that, only reception can.	2 hours	BKG-05, BKG-06
BR-05	**Check-in window, grace and late arrival.** Check-in opens 60 minutes before the slot. A patient who checks in by slot start + grace period keeps appointment priority. A patient who checks in later, but before the no-show limit (slot start + 60 minutes), is a late arrival: they keep the appointment, join the queue from their check-in time (BR-08) and do not count as a no-show.	60 min / 15 min / 60 min	QUE-01, QUE-02
BR-06	**Automatic no-show.** A Booked appointment not checked in by the no-show limit (slot start + 60 minutes) becomes No-Show. The queue is not affected. Reception can reverse a No-Show on the same day, with a reason; the reversal is logged.	60 min	QUE-14
BR-07	**Repeat no-shows.** After 3 no-shows in 60 days, online booking is disabled for 14 days. Reception can still book and the admin can lift the restriction.	3 / 60 days / 14 days	ADM-04
BR-08	**Queue order.** Sort by (1) emergency first, then (2) effective time, earliest first, then (3) check-in time, earliest first. Effective time is the slot start for appointment patients who check in on time, and the check-in time for walk-ins and late arrivals.	—	QUE-01, QUE-03, QUE-06
BR-09	**Emergency priority.** Only a receptionist or doctor can set an emergency flag, and a reason is required. If there are several, they are ordered by the time they were flagged.	—	QUE-09
BR-10	**Skip rule.** A doctor can skip a called patient who does not respond. The patient returns to Waiting with effective time = now (behind all current waiters). After 2 skips, only reception can re-add or remove them.	2 skips	QUE-08, QUE-10

ID	Rule	Default	Stories
BR-11	**One at a time.** A doctor can have at most one patient In Consultation. The doctor must complete or skip that patient before calling the next.	—	QUE-07
BR-12	**Estimated waiting time.** EWT = (patients ahead $\times$ doctor's average consultation time) $-$ time already spent with the current patient (never below 0). The average uses the doctor's last 20 completed consultations; the default is the slot length when fewer than 5 exist. It is shown as a range of -20% to $+20\%$, rounded to the nearest 5 minutes.	20 visits / 5 min	QUE-04, QUE-13
BR-13	**Doctor leave.** When leave is saved, no new slots are offered for that period. Existing appointments in the period become Needs Rescheduling and the patients are notified. Reception can help rebook.	—	SCH-03, NOT-03
BR-14	**Token numbers.** Tokens run 001, 002, … per doctor per day and restart each day. They are shown with the doctor code (for example D2-014).	—	QUE-01
BR-15	**Clinical notes.** Only the authoring doctor can edit a note, and only within 24 hours. Later changes are stored as visible amendments and the original is kept. Who can read notes is defined in PRIV-05.	24 hours	VIS-01, VIS-02, VIS-07
BR-16	**Upcoming appointments cap.** A person can hold at most 3 upcoming active appointments.	3	BKG-07

Appointment lifecycle

Every appointment has one status at a time. The full state diagram will be drawn in the SDS; this table is the agreed starting point.

Status	Meaning	Can change to
Booked	Reserved; the patient has not yet arrived.	Checked-In, Cancelled, No-Show, Needs Rescheduling
Needs Rescheduling	The doctor is on leave; a new time is needed.	Booked (new slot, by reschedule or through reception), Cancelled
Checked-In	The patient has arrived (on time or late) and is in the queue.	In Consultation, Cancelled (left the queue)
In Consultation	The doctor is seeing the patient.	Completed
Completed	The consultation is finished.	Final
Cancelled	Cancelled by the patient or by reception.	Final
No-Show	The patient did not check in by the no-show limit.	Checked-In (only by reception, same day, with a reason); otherwise final

Appointment types: Online, Reception, Walk-in (starts at Checked-In), Follow-up. **Queue entry states:** Waiting, Called (In Consultation), Skipped (returns to Waiting), Done, Removed.

3.4 External Interface Requirements

Interface	Description	Requirements
User interface	Responsive web app with role-based screens: patient (search, book, my appointments, queue, records), receptionist (booking, check-in, walk-in, queue), doctor (schedule, queue, visit), admin (staff, rules, reports, audit).	USAB-01 to USAB-06, COMP-01, COMP-02
Waiting-room display	A read-only public screen with "Now serving" tokens per doctor.	QUE-11, QUE-15, PRIV-03, COMP-03, USAB-07

Interface	Description	Requirements
Hardware	Reception PC, doctor PC or tablet, waiting-room screen, printer for slips and prescriptions, phone camera for QR scan (optional).	A-07, QUE-02, VIS-03
SMS gateway	Sends OTP codes and notifications through a provider's web interface. Wrapped in a replaceable adapter so the provider can change. A sandbox or simulator is used during development.	AUTH-01, NOT-01 to NOT-05, REL-04
Email service	Sends password resets and notifications (SMTP or provider API).	AUTH-06, REL-04
Database	Stores all data described in Section 3.5.	REL-02, REL-03, SEC-07
Communication	HTTPS. Live updates by server push (for example WebSocket or server-sent events) or, as a fallback, by polling at most every 2 seconds, so that the update delay plus processing stays within the 5 seconds of PERF-03.	SEC-03, PERF-03

3.5 Data Requirements

These are the main pieces of information the system must keep. They will be refined into data flow diagrams and a database design in the SDS.

Data item	Purpose	Key information
Patient	The person receiving care	Name, date of birth or age, gender, mobile number, address, emergency contact, language and notification choice, restriction status
Dependent	A person managed by a guardian	Guardian patient, name, date of birth, gender
Staff user	Doctor, receptionist or admin login	Name, email, role, status
Doctor	Doctor details for booking	Specialty, room, photo, slot length, active status
Schedule rule	Weekly availability	Doctor, weekday, start and end time, breaks
Leave / closure	Doctor leave and clinic closed days	Doctor or clinic, date range, reason
Slot	A bookable time block	Doctor, start and end, state (free, booked, blocked)
Appointment	A booking	Reference number, person, doctor, slot, type, status, reason, who changed it and when
Queue entry	A patient's place in a doctor's daily queue	Appointment, token, effective time, check-in time, emergency flag, state, skip count
Visit record	What happened in the consultation	Appointment, doctor, notes, advice, vitals, amendments
Prescription line	One medicine on a prescription	Visit, medicine, dose, frequency, duration
Notification	A message to a user	Recipient, channel, content, status, attempts
Audit event	A record of a sensitive action	Who, role, action, item, time
Data request	A patient's request for a copy of their data or for deletion (SYS-05)	Patient, type, status, date requested, handled by, date handled
Clinic settings	Rule values (Section 3.3)	Booking window, cut-off, grace period, limits
Waiting-list entry (Could)	A request for a freed slot	Patient, doctor, date, time notified

Appendix A – Traceability and Coverage

A.1 Stories by user role and epic

This table shows who each epic serves. “Staff” means stories open to any staff member. Every role has stories, and the three primary users all have Must stories.

Epic	Patient	Reception.	Doctor	Admin	Staff	Total
AUTH	5	1	—	2	1	9
SCH	—	1	2	3	—	6
BKG	9	2	—	—	—	11
QUE	6	5	4	—	—	15
VIS	1	1	5	—	—	7
NOT	5	—	—	—	—	5
ADM	—	—	—	4	—	4
SYS	3	1	—	1	—	5
Total	**29**	**11**	**11**	**10**	**1**	**62**

A.2 Business rules to stories

Rule	Stories that apply it	Unit test needed (when)
BR-01	SCH-02, BKG-03	Yes (MAIN-02), Release 1
BR-02	BKG-07	Yes (MAIN-02), when built (Release 2)
BR-03	BKG-02, BKG-03	Yes (MAIN-02), Release 1
BR-04	BKG-05, BKG-06	Yes (MAIN-02), Release 1
BR-05	QUE-01, QUE-02	Yes (MAIN-02), Release 1
BR-06	QUE-14	Yes (MAIN-02), when built (Release 2)
BR-07	ADM-04	Yes (MAIN-02), when built (Release 3)
BR-08	QUE-01, QUE-03, QUE-06	Yes (MAIN-02), Release 1
BR-09	QUE-09	Yes (MAIN-02), Release 1
BR-10	QUE-08, QUE-10	Yes (MAIN-02), Release 1
BR-11	QUE-07	Yes (MAIN-02), Release 1
BR-12	QUE-04, QUE-13	Yes (MAIN-02), Release 1
BR-13	SCH-03, NOT-03	Yes (MAIN-02), Release 1
BR-14	QUE-01	Yes (MAIN-02), Release 1
BR-15	VIS-01, VIS-02, VIS-07	Yes (MAIN-02), Release 1
BR-16	BKG-07	Yes (MAIN-02), when built (Release 2)

A.3 Risks to requirements

Each risk in Section 1.5 lists the stories, rules and non-functional requirements that reduce it. The table below shows when each High risk first gets a mitigation. A “shall” requirement in Section 3.2 is mandatory in every release.

Risk	First release with a mitigation	Mitigation and note
R-01	Release 1	BKG-03 (Must), BR-01, REL-02.
R-03	Release 1	QUE-09 (Must), SAFE-02.
R-04	Release 1	SEC-01, SEC-06 and PRIV-05 are “shall” requirements. AUTH-07, VIS-02, AUTH-08 and AUTH-09 follow in Release 2.
R-06	Release 1	NOT-01 (Must), REL-04.
R-08	Release 1	QUE-04 shows a range (Must) and PERF-05 sets the accuracy target; QUE-13 follows in Release 2.
R-09	Release 1	SYS-04 (Must) gives a printed schedule; REL-01 sets the availability target.
R-11	Release 2	Accepted for the Release 1 demo (one clinic, synthetic data). BKG-07, QUE-14 and ADM-04 mitigate it from Release 2. The team should confirm this decision.

A.4 Where this SRS feeds the SDS

SDS artefact	Taken from this SRS
Use case diagram	The four roles in 2.1 as actors, and the Must and Should stories in 3.1 as use cases.
Data flow diagram	The data items in 3.5 and the interfaces in 3.4 (patient, staff, SMS gateway, email, display).
State diagram	The appointment lifecycle and queue entry states in 3.3.
Architecture (MVC / ECB / CQRS)	The constraint C-04 and requirement MAIN-01; reads and writes split naturally between search and queue views and booking commands.

A.5 Dependency check

Rule: the confirmation criteria of a Must story must not need a Should or Could story. The review of version 1.0 found these cases and resolved them.

Must story	Needed	How it was resolved
QUE-07	QUE-11 (display) and NOT-04 (phone alert)	The criterion now uses the patient’s live queue screen (QUE-04). The display and the alert stay in their own Should stories.
SCH-03	NOT-03 (leave notice)	NOT-03 was raised to Must (3 points).
QUE-08	QUE-10 (move or remove)	QUE-10 was raised to Must (3 points) because BR-10 needs it after 2 skips.
NOT-03	BKG-06 (one-step reschedule)	The link opens the appointment, where the patient can cancel and book again. One-step reschedule stays Should.
SYS-04 (risk R-09)	—	Raised to Must (2 points) so that the highest-rated availability risk has a Must mitigation.

Appendix B – Release Plan

The plan follows the timebox idea from Lecture 2: the time is fixed and the scope flexes. Release 1 is the minimum that tells the complete demo story (book, check in, queue, consult).

Release	Contents	Stories	Points
Release 1 (core)	All Must stories	26	109
Release 2	All Should stories, in priority order, as time allows	28	101
Release 3	Could stories if time remains	8	24

> **Capacity check.** Constraint C-01 gives about 9 one-week sprints (weeks 3 to 11). Release 1 needs six of them (109 points at a planning velocity of 20, with Sprint 6 kept light). About three sprints, roughly 60 points, remain, so Release 2 (101 points) is a prioritised backlog and not a commitment. Suggested order for the first Release 2 stories (57 points): BKG-07, QUE-14, NOT-04, QUE-11, NOT-02, QUE-13, BKG-06, QUE-05, AUTH-07, VIS-02, SYS-05, AUTH-08, SYS-03, VIS-05, VIS-03.

Suggested sprints for Release 1

Assumption: one-week sprints and a planning velocity of 20 points. The team must replace this with its measured velocity after Sprint 1. If velocity is lower, the scope is reduced (a story moves down a priority) and the date is not moved.

Sprint	Goal	Stories	Points
Sprint 1	Accounts and doctor set-up	AUTH-01, AUTH-02, AUTH-05, AUTH-06, SCH-01	19
Sprint 2	Schedules and finding slots	SCH-02, SCH-03, BKG-01, BKG-02, SYS-02	19
Sprint 3	Booking	BKG-03, BKG-04, BKG-05, BKG-09	19
Sprint 4	Check-in and doctor queue	QUE-01, QUE-03, QUE-06, QUE-07	20
Sprint 5	Live estimates, completion, emergencies, notes	QUE-04, QUE-08, QUE-09, VIS-01	21
Sprint 6	Confirmations, leave notices, queue fixes, printed schedule, bug fixing and demo preparation	NOT-01, NOT-03, QUE-10, SYS-04	11
Total			**109**

Sprint 6 is deliberately light (11 points) so that the team has room for bug fixing, the first Should stories, the report and demo preparation. It also completes the confirmation criteria of BKG-03, BKG-05 and SCH-03 that use notifications (NOT-01, NOT-03). Sprint 0 (before Sprint 1) covers the SRS, SDS, repository, Trello board and Wiki set-up.

Appendix C – Elicitation and Validation Log

C.1 How these requirements were developed

This draft was produced with the help of a generative AI assistant, working from the course material, and was then reviewed by the team. Personas, examples and numbers are fictional or assumed until they are validated (C.2). The following techniques were used:

- **Study of the course material:** SRS guide, project assessment and lectures, to match the required structure, user-story-with-confirmation format and quality items.
- **Personas and journey mapping:** five personas and the seven-stage visit journey (Section 2) to find what each user needs at each moment.
- **Role by data brainstorming:** for each data item (appointment, queue entry, visit, schedule, account) and each role, we asked: who creates, views, changes and removes it? This produced the epics and many stories.
- **"What could go wrong" brainstorming:** double booking, unfair queue, emergencies, privacy, outages and abuse produced the risk register (1.5), the business rules (3.3) and the safety and security requirements.
- **Quality checks:** each story was checked against INVEST, given testable confirmation criteria, a MoSCoW priority and a size estimate.
- **Standards check:** non-functional requirements follow ISO/IEC 25010 categories with measurable targets; document structure follows ISO/IEC/IEEE 29148.
- **Consistency review (version 1.1):** stories, business rules, non-functional requirements, risks and the release plan were cross-checked against each other; the changes are listed in the revision history.

C.2 Validation still to be done by the team

Requirements are not finished until real users have reviewed them. The team should complete this table, then update the SRS version and revision history. Do not invent entries: record only what actually happened.

Activity	With whom	Date	What we learned / changed
Interview about phone calls, walk-ins and the paper register	Receptionist (at least 1)		
Interview about schedules, queue and notes	Doctor (at least 1)		
Interview about booking and waiting experience	Patients (at least 5)		
Review of rules, reports and staff access	Clinic manager or owner		
Walk-through of paper wireframes against the stories	Any 3 stakeholders		
Review of this SRS	Course instructor		

Appendix D – Definition of Ready and Done

Definition of Ready (a story may enter a sprint when…)

- It follows the story format and passes the INVEST checks.
- It has at least one confirmation criterion that can be tested.
- It has a point estimate of 8 or less.
- Its dependencies (other stories, rules, data) are known.
- Its confirmation criteria do not depend on a story of lower priority (Appendix A.5).
- A UI sketch exists if the story has a screen.

Definition of Done (a story is finished when…)

- The code is on a feature branch and merged by a pull request with at least one peer review (MAIN-04).
- Unit tests for the story and its business rules are written and pass in CI (MAIN-02).
- Code is documented and follows the coding standard (MAIN-03).
- Each confirmation criterion has been demonstrated successfully.
- The screens were checked against Nielsen’s heuristics (USAB-01).
- The GitHub Wiki and Trello card are updated.
- No known High-severity bug remains.